

CS610 (COMPUTER NETWORKS) MID TERM QUESTION ANSWER BY MOHSIN RAZA
03137248280 hussainzakki5@gmail.com

Q 1: Where are destination and source address located in frame?

The packet header includes destination and source addresses.

Q 2: If 100mbps Ethernet NIC is connected to a hub which supports 10mbps, what will be the data rate in this scenario?

This has two results. First, the hub does not necessarily create a broadcast medium. Second, there are no collisions between traffic on 10 Mbps ports and 100 Mbps ports.

Q 3: In connection oriented network which type of connection identifier is used? Support your answer with example.

ATM uses the concept of connection-oriented networking.

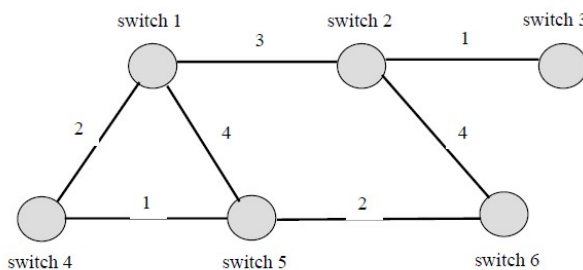
Q 4: Imagine a network of 6 devices A, B, C, D, E and F. All devices are on same packet switch. If computer A wants to send packet to computer E, How would packet switch forward this packet?

Path between switch A and F and the total distance traveled along this path

Switch A – switch D – switch E

P 1

Distance = 4



Q 5: Network administrator needs audio and video service for his network. He has to send or receive data at same data rate. Which type of ATM quality service specification would be applied there? Give example?

To make possible the transport of audio and video data across Asynchronous Transfer Mode (ATM) networks, a simple, low cost, audio/video ATM appliance, the AVATAR, has been developed. This appliance is capable to handling uncompressed bidirectional audio and NTSC video connections.

Q 6: We have two satellite locations, at first location one bridge is configured and at second location 2 bridges are configured. Which location performed will be faster? Give reason.

Satellite locations with two bridges configured will performed faster. The brigades also provide power virtually to its East Timor assets. A bridge that has multiple ports is known as a networking switch. Both bridges and switches are capable of directing traffic to specific network addresses instead of broadcasting the data to all devices on a network segment. This functionality makes the bridge or switches a more advanced networking device over a hub or repeater.

Q 7: There are two sites office A & B, both are using VPN. If a user of another network wants to send some packet to A, then what will be the behavior of network A? How data is protected in VPN environment?

When private network “A” wants to send a packet to a private network “B” the packet is accepted by the VPN server. Then a standard IP packet that has a destination address corresponding to the VPN server at network B. The machines within network A and B don't have publicly accessible address – Because they not on the public Internet. This is advantage of the VPN approach.

Q 8: Star Textile has huge network of systems and switches. A task is given to IT Officer that if some link goes the entire network works smoothly. Which routing method will be use?

Distributed routing method is better for this purpose because Distributed routing build the required connections with its neighbors. Distributed routing is more flexible than centralized routing because each node handles its own routing. The result is regularly improved system performance.

Q 9: Network administrator wants to build a tunnel between sites offices. How will he build using private virtual network?

To establishes a private network that can send data securely between these two locations or networks through a tunnel. A VPN tunnel connects the two PCs or networks and allows data to be transmitted.

Q 10: Who examine performance of network?

Performance is examined by users in the network environment.

Q 11: How can we compute shortest path in WAN?

We use Djikstra’s algorithm to compute shortest path from each node to every other node.

P 2

Q 12: What type of switching technique is used in WAN?

WANs can use circuit-switching or packet-switching techniques. Each switch moves an entire packet from one connection to another. That's why they are called packet switches.

Q 13: Write the names of identifiers used in ATM.

An ATM with 16-bit address, 24-bit connection identifier and connection identifier includes.

8-bit Virtual Path Identifier (VPI)

16-bit Virtual Circuit Identifier (VCI)

Q 14: What type of topology used in Ethernet?

Traditional Ethernet use a bus topology, meaning that all devices or hosts on the network use the same shared communication line.

Q 15: Difference between LAN and WAN.

In LAN (Local Area Network) network occupies the smaller area like a room or a building. In WAN (Wide Area Network) network occupies larger areas like cities & countries. Internet is a Wide Area Network.

Q 16: Why fiber need for ATM?

P 3

ATM is designed to work on fiber (but can be used with twisted pair). A typical port on an ATM switch operates at OC-3 speed (155Mbps) or higher.

Q 17: Define Bridge.

A bridge is a hardware device also used to connect two LAN segments to extend a LAN. A typical bridge has two NICs. It only runs the code stored in its ROM.

Q 18: Ambiguity may cause the failure of any network, how parity bit check is consider in this situation.

Parity can only detect errors that change in odd number of bits. We see that even no. of bits have been changed due to noise so parity checking cannot detect this error.

Q 19: Define default route how it is different from routing.

Routing table entries can be twisted with a default route. If the destination does not have an explicit routing table entry, then it uses a default route. The process of forwarding the packets of information is called routing. The information about destinations is kept in routing tables.

Q 20: Defines simplex and full duplex connection?

Some connection-oriented technologies provide full duplex while other allow on simplex connection. To communicate using a simplex connection a pair of computers must establish two connections one from computer A to computer B and another from computer B to A.

Q 21: How WAN capable to handle many computers?

A Wide Area Network (WAN) is a telecommunication network that covers a broad area.

Q 22: Briefly explains the working structure of thick Ethernet.

It uses thick coax cable. AUI cable (or transceiver or drop cable) connects from NIC to transceiver. AUI cable carries digital signal from NIC to transceiver. The wires in AUI carry digital signals power and other control signals. Thick Ethernet also requires terminators to avoid signal reflectance.

Q 23: Why ATM design chose cells over packet?

Cells are not variable length and memory management. With cells bits can be counted as they arrive.

Q 24: What is 10base T?

The 10Base-T standard (also called Twisted Pair Ethernet) uses a twisted-pair cable with maximum lengths of 100 meters. This is another standard of wiring scheme. 10Base-T Ethernet is often called a star shaped bus.

Q 25: What is Multicasting?

It works like broadcasting but it does not forward frames automatically to the CPU.

Q 26: What is FDDI?

Fiber distributed data interconnect (FDDI) is another ring technology. Its important features are: It use fiber optics between stations and transmits data at 100Mbps. It uses pair of fibers to form two rings.

Q 27: Define 802.11 Wireless LAN and CSMA?

IEEE 802.11 is standard wireless LAN that use radio signals at 2.4GHz. Its data rate is 11Mbps. The older devices use radio signals at 900MHz and data rate of 2Mbps. Bluetooth specifies a wireless LAN for short distances.

Q 28: Define static and dynamic routing.

P 4

Static Routing:-

Static routing is the simplest form of routing, but it is a manual process.

Dynamic Routing:-

Dynamic routing protocols are supported by software applications running on the routing device (the router).

Q 29: Disadvantages of Repeaters.

Repeaters do not recognize frame formats; they just amplify and retransmit the electrical signal. If a collision or error occurs in one segment, repeaters amplify and retransmit also the error to the other segments.

Q 30: Different b/w Cells and Packets?

ATM designers chose cells over packets because of the following reasons:
Cells are not variable length and memory management. Handling variable length packets leads to memory breakup. With cells bits can be counted as they arrive.

Q 31: What is the difference between the physical and logical topologies?

Physical Topology:

The way that the workstations are connected to the network through the actual cables that transmits data. The physical structure of the network is called the physical topology. It depends on the wiring scheme.

Logical Topology:

The logical topology is the way that the data passes through the network from one device to another without the physical interconnection of the devices. We can say that it is defined by the specific network technology.

Q 32: What is meant by Bridges STARTUP and STEADY State?

When a bridge boots the address lists be empty (start up state). The bridge forwards frames to the other segment if it cannot find destination address in its lists. After some time when the bridge received at least one frame from every computer, it has the lists built (steady state).

Q 33: How can “Switched Virtual Network” are established? P 5 ESTABLISHING AN SVC (switched virtual circuit):

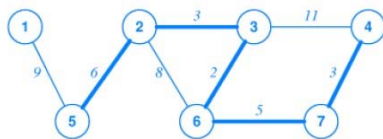
The computer sends a connection request to the switch attached with it. Software in the switch finds a network path to the destination and sends along the connection request. Once the connection is established by the destination, a message is sent back to the computer to indicate the SVC is set. If any switch or the destination computer does not agree to setting up the VC, an error message is sent back and the SVC is not established.

Q 34: What is the concept of packet switches?

Each switch moves an entire packet from one connection to another. That's why they are called packet switches. Packet switches consist of a small computer with network interfaces.

Q 35: Write a note on Weight Graph.

Dijkstra's algorithm can contain weights on edges in graph. The shortest path is the path with lowest total weight (sum of the weight with all edges). For example as shown in the figure below:



The shortest path in the figure from node 2 to node 6 is 2 to 3 and 3 to 6 as this path has the smallest weight so it is the shortest path.

Q 36: Define Vector-Distance Algorithm.

Packet switches wait for next update message and they iterate through entries in message. If entry has shortest path to destination, insert source as next step to destination and record distance.

Q 37: What is the concept of store and forward technology?

STORE AND FORWARD: Data delivery from one computer to another is accomplished through store and forward technology. In this technology packet switch stores incoming packet and also forwards that packet to another switch or computer.

Q 38: How can a bridge know whether to forward frames?

The bridges organize themselves automatically to decide which bridge will forward frames and which bridge will not. The bridges communicate with each other on the network and use Distributed Spanning Tree (DST) algorithm to decide which bridge will not forward frames if a cycle occurs.

Q 39: Compare connection oriented and connectionless Service.

In connection-oriented protocol, authentication is needed while this is not in case of connectionless protocol.

In connection-oriented protocol, we have to establish connection between sender and receiver while this is not in case of connectionless protocol.

Example of connection-oriented protocol is TCP and the example of connectionless protocol is UDP Internet.

TCP is a connection-oriented protocol, it makes a connection and checks whether the data is received, and resends if it is not. UDP is a connectionless protocol; it has not guarantee delivery.

Q 40: Give a comparison of wiring Schemes.

P 6

The wiring schemes are compared as follows:

Separate transceiver allows computers to be powered off or disconnected from network without troublemaking. Transceiver may be located in an awkward place, so finely fault transceiver can be hard. In other case, thin coax cable takes minimum of cable. Disconnecting one computer (on one loose connection) can disturb entire network. Hub wiring merges electronics and connections. 10Base-T is most popular because of lowest cost.

Q 41: Describe permanent virtual circuits (PVC).

ATM can provide customers with virtual circuits that look like traditional digital circuits. The forwarding tables are automatically restored after breakdown power of equipment. The forwarding table entries for such permanent VC's are statically configured.

Q 42: What are default routes draw the table.

Routing table entries can be twisted with a default route. If the destination does not have in routing table entry, then it uses a default route. Default routes for 4 nodes are shown in the figure below.

destin- ation	next hop	destin- ation	next hop	destin- ation	next hop	destin- ation	next hop
1	-	2	-	1	(3,1)	2	(4,2)
*	(1,3)	4	(2,4)	2	(3,2)	4	-
		*	(2,3)	3	-	*	(4,3)
				4	(3,4)		
node 1		node 2		node 3		node 4	

Q 43: Star Organization use four bridges in its network and they form a cycle. How can network engineer avoid broadcasting from all bridges in this scenario?

If a bridge network forms a cycle, then not all bridges on the network are allowed to forwarding broadcasting frames. The bridges configure themselves automatically to decide which bridge will forward broadcast frames and which bridge will not. The bridges communicate with each other on the network and use Distributed Spanning Tree (DST) algorithm to decide which bridge will not forward frames if a cycle occurs.

Q 44: You are working in a star Organization as Network Engineer. The Existing Network comprises of 120 Systems. What will be your analysis about delay shout it be Smaller or higher?

The computers take turns using the medium. First one computer uses the medium to send its data over the channel then second and so on. But sharing a single medium over long distances is efficient, due to the long delays. The computer with shorter delay will go first and other computer may transmit later.

Q 45: We want to connect two site Offices, Which are far away from each other. In your opinion, which one is best LAN extensions or WAN? Justify your answer.

WAN extension is best.

Reasons:

P 7

1. It integrate IT infrastructure
2. It boosts your privacy
3. It increases bandwidth

Q 46: Suppose ABC Corporation has a small number of packet switches and Systems. The entire works smoothly, for this type of Setup which routing method is most appropriate? Give Reason.

STATIC ROUTING:

It is done at boot time. It is simple and has low network overhead. It is inflexible.

Q 47: Explain Classification of Networks?

Computer networks are classified by four factors which are as follow:

1) BY SIZE:

- 2) BY CONNECTIVITY:
- 3) BY MEDIUM:
- 4) BY MOBILITY:

BY SIZE:

According to their size there are two classifications of networks.

1. Local Area Network. (LAN)
2. Wide Area Network (WAN) **BY CONNECTIVITY:**

Networks are also classified by connectivity in two topologies. a) Point-to-Point

b) Broadcast

BY MEDIUM:

The classification of networks is also based on the Medium.

- Copper wire
- Co-axial cable
- Optical fiber
- Radio waves

BY MOBILITY:

The networks are also classified according to their mobility. In this respect there are two types of networks.

- Fixed networks
- Mobile networks

P 8

Q 48: Define network topology and VPN?

10Base-T network topology is a bus but wiring topology is a star. There are three most popular topologies. • Star • Ring • Bus

STAR TOPOLOGY: In this topology, all computers are attached to a central point, which is sometimes called the “Hub”

RING TOPOLOGY: In this topology of network the computers are connected to each other in closed ring. In this network first computer passes data to the second and then second passes data to third and so on,

BUS TOPOLOGY: In a bus topology all computers are attached to a single long cable and any computer can send data to any other computer.

Virtual Private Network (VPN) combines the features of both private and public networks. It is limited to single organization and uses public network for connectivity.

Q 49: Suppose you are assigned to connect two site offices which are situated far away from each other. One office in Lahore and other is in Paris. In your opinion which one will be the appropriate solution for this scenario LAN extension or WAN?

Justify your answer with reasons.

Answer:

WIDE AREA NETWORK (WAN):

It is used for a country level networking and even for continents. In WAN, network occupies larger areas like cities & countries.

Q 50: Suppose you want to add large amount of systems in your WAN to increase its capacity. Which type of switching devices will be used? Also write the names of these switching devices along with difference between them.

We can add capacity to WAN by adding more links and packet switches. WANs can use circuit-switching or packet-switching techniques. Each switch moves an entire packet from one connection to another. That's why they are called packet switches.

Q 51: Which approach of error detection is able to detect more errors as compared to others, considering that there should be no increase in the size of information being transmitted? Explain how it works.

CRC has good error detection properties. It is easy to implement in hardware. To enable a network system to detect more error without increasing the amount of information in each packet is called CRC.

Q 52: During data transmission, what are the possible indications if, a)

A frame is missing SOH on receiver side

b) A frame is missing EOT on receiver side

Missing SOH indicates receiving computer missed beginning of message. Missing EOT indicates sending computer crashed.

Q 53: Network manager wants absolute privacy in his organization's computer networks even using public network. Which technology will be used and how it assures absolute privacy of packets?

In addition to check packets, VPN systems use encryption to guarantee absolute privacy.

Q 54: Suppose an organization wants to implement ATM technology in its network. Which type of connection ATM technology will provide in network?

ATM uses the concept of connection-oriented networking. Connections in ATM are called virtual channels (VC) or virtual circuits (a term preferred by some). An ATM network is built from interconnected ATM switches. The attachment points or ports can be connected to computers or other ATM switches.

Q 55: Suppose in computer network of a chemical company, each packet is transferred with equal amount of time between systems. Considering this scenario what will be the jitter of the network.

A network with zero jitter takes exactly the same amount of time to transfer each packet. A network with high jitter takes much longer to deliver some packets

Q 56: Write the name (Abbreviation) of ATM?

ATM stands for (ASYNCHRONOUS TRANSFER MODE).

Q 57: How to increase the capacity of WAN?

We can add capacity to WAN by adding more links and packet switches.

Q 58: We have to use repeaters to extend LAN. Is there any problem in LAN due to repeaters? Give Reason.

A bridge is a hardware device also used to connect two LAN segments to extend a LAN. Unlike a repeater, a bridge uses two NICs to connect two segments. It listens to all traffic and recognizes frame format. It also forwards only correct complete frames and discards the crash and error frames.

Q 59: Define two types of Virtual Channel or circuit used in ATM?

VPI or Virtual Path Indicator (8- bit),

VCI or Virtual Channel Indicator (16-bits),

Q 60: Which of the following are Ethernet fields?

CRC, Flag, Frame Type, Preamble, Destination Address, Source Address and Data,

<u>FIELD</u>	<u>PURPOSE</u>
Preamble	Receiver synchronization
Destination Address	Identifies intended receiver
Source Address	Hardware address of sender
Frame Type	Type of data carried in frame
Data	Frame payload
CRC	32-bit CRC code

Q 61: As we know when bridge network forms a cycle then not all bridges on the network must be allowed to forward broadcast frames So tell the which algorithm resolve this problem?

The bridges communicate with each other on the network and use Distributed Spanning Tree (DST) algorithm to decide which bridge will not forward frames if a cycle occurs.

Q 62: Which technology is used to copy frame data directly from main memory?

LAN interface may use DMA technology to copy frame data directly from main memory.

Q 63: ATM network specifies a 24 bits VPI/VCI connection identifier. How many bits VPI and VCI occupies individually?

VPI or Virtual Path Indicator (8- bit),

VCI or Virtual Channel Indicator (16-bits),

Q 64: Why repeaters are not used for extension with large networks? Give at least two reasons.

Unlike a repeater, a bridge uses two NICs to connect two segments. It listens to all traffic and recognizes frame format. It also forwards only correct complete frames and discards the crash and error frames.

Q 65: Suppose in a university campus, bridges are used in its local area network. Explain all steps, how bridge builds up address list table of all attached computers?

A bridge keeps a list for segment that consist physical addresses of the computer attached to that segment. In this way a bridge knows on which segment a destination computer is attached. Most bridges are self learning bridges.

When it boots its address table is empty. When it receives frames from other computers it starts building up its address table.

Q 66: Suppose we have to extend an Ethernet segment up to three segments then how many repeater will be needed for this purpose?

One repeater doubles, two repeaters triple the maximum cable length limitation.

Computers attached to different segments communicate as if they are connected to the same cable.

Q 67: Assume ABC Company has two LAN segment connected through bridge. In which state both LAN segment can communicate simultaneously?

In a steady state, a bridge allows simultaneous use of segment and then both LAN segment can communicate simultaneously in this state.

Q 68: Is it necessary to use terminator at both ends of thin Ethernet cable? Give your answer with the solid reason.

There is a lot of signal reflectance in the network. To avoid this we use terminators at the end of Thin Ethernet

Q 69: A specialized computer system is used in a network in order to measure the performance of network what is specific name for such computer network and mention any three type of information it can provided in this regard?

A network analyzer also called network monitor is used to examine the performance of a network. It can report information such as capacity utilization, distribution of frame size, collision rate or token circulation time.

Q 70: Define three benefits of ATM.

ATM is a single technology that is designed to meet the goals of both LANs and WANs.

ATM uses the concept of connection-oriented networking.

It transmits data at over 100Mbps.

It uses fiber optics to connect computer to switch.

Each connection includes two fibers.

Q 71: What is the difference between LAN and WAN? Local area network (LAN)

LAN is small in size covers the area within a room, building or cities. **Wide area network (WAN)**

WAN is large in size and covers the area cities, countries and continents.

Q 72: Define the term Jitter.

The term Jitter is difference in transmission delays. Jitter is significance for voice, video and data. Jitter can occur when a packet is delayed because the network is busy.

Q 73: Briefly Explain Thick Ethernet wiring scheme.

THICK ETHERNET WIRING:

It uses thick coax cable. AUI cable (or transceiver or drop cable) connects from NIC to transceiver. AUI cable carries digital signal from NIC to transceiver. It requires terminators to avoid signal reflectance.

Q 74: If we have 8 computers which type of Ethernet is used thick or thin? Thin

Ethernet is useful when many computers are located close to each other.

Q 75: Give Some Advantage of bridge.

A bridge is a hardware device also used to connect two LAN segments to extend a LAN. Most bridges are self learning bridges.

A bridges network can connect many segments.

The bridges configure themselves automatically to decide which bridge will forward broadcast frames and which bridge will not.

P 12

Q 76: Write Types of delay.

There are following types of delay:

PROPAGATION DELAY: It defined as the time to travel across medium.

SWITCHING DELAY: It is the time required for network component (hub, bridge, packet switch) to forward data.

ACCESS DELAY: It is the time required to get control of medium (CSMA/CD, token).

QUEUEING DELAY: It is the time enquired in packet switches.

Q 77: Suppose a network administrator wants to get automatic updates of routing information in a router, if any changes occur in the network, which routing algorithm will be used in this case? Give reason.

DYNAMIC ROUTING: It allows automatic updates to a programmer. It can work around network failures automatically.

Q 78: Routing of packet from source to destination does not depend on source address of a packet write any two advantages of this In-dependency.

It has several benefits. It allows fast and efficient routing.
The network can continue to function even if topology changes without notifying entire network.

Q 79: Define QoS services briefly.

There are three types of QoS specifications, which are given as follows:

CONSTANT BIT RATE (CBR):

It is used for audio and video, since these have predefined maximum data rates.

VARIABLE BIT RATE (VBR):

It is used for compressed audio and video where the data rate depends on the level of compression.

AVAILABLE BIT RATE (ABR):

It is used for typical data applications (where the data rate may be unknown) and allows use of whatever bandwidth is available at a given time.

Q 80: Which type of quality service specifies of ATM would be applied in each of the following?

- (a) When network engineer want to use ATM for compressed audio and video.....CBR
- (b) Network administrator needs services for typical data application.....ABR

Q 81: Suppose a network administration of XYZ organization given a task to develop a network in a limited area with low cost then which wiring scheme will you use? Justify your answer with reasons?

Twisted pair Ethernet is the cheapest wiring that makes it so popular.

Q 82: Describe the structure of physical address in WAN if you have (1, 5) on a switch then who will you elaborate this address?

Many WANs use hierarchical addressing for efficiency. One part of address identifies destination switch. Other part of address identifies port on switch.

P 13

Q 83: What is best in CBR VBR or ABR?

ABR (Available Bit Rate) is best, because it is used for typical data applications and allows use of whatever bandwidth is available at a given time.

Q 84: What service specification would be used when intend to transmit data which is less critical.

Variable Bit Rate (VBR) And Available Bit Rate (ABR) both can be implied.

Q 85: What is the solution to avoid duplication of redundant information in address table?

Routing table entries can be bent with a default route. If the destination is not clear in routing table entry, then it uses a default route.

Q 86: Write down the names of switches used to add many systems to WAN. Also differentiate these switches

Interior switch: The switch that has no attached computers is called an interior switch.

Exterior switch: The switch that has computers attached with it is called exterior switch. Both interior and exterior switches forward packets and they also need routing tables.

Q 87: Explain how does a bridge build up its address table?

When it boots its address table is empty. When it receives frames from other computers it starts building up its address table. The table is fully occupied once it had received at least one frame from attached computer.

Q 88: Can we increase the maximum cable length as many times as we wish by just adding repeaters?

No

Q 89: Due to Which reason ATM is adopted as single underline technology.

It was designed as a single technology for voice, video and data and has low jitter (difference in delivery time) and high capacity.

Q 90: Which type of wireless use Wi-Fi?

P 14

See From Google.



